IAN HILL

Phone +1 (778) 580-6694
Email ianrmhill@gmail.com
Website ianhill.ca
Publications orcid.org/0000-0001-9782-8522
Repositories github.com/ianrmhill

Key Qualifications

> Forefront contributor to semiconductor reliability research including monitoring, prediction, test, and statistical methods.

> Expertise in probabilistic programming paradigms for generative modelling, inference, and experimental design.

> Adept research skillset spanning problem identification, methods ideation, statistical results analysis, and authorship.

> Experienced in a majority of ASIC design and verification topics across both analog and digital domains.

> Versatile software, hardware, and ancillary tools proficiency to tackle broad-ranging engineering projects.

> Strong technical leadership skills applicable to research lab and project management roles.

Research Summary

Completing Ph.D. in 2025, currently with 5 first-author IEEE publications (+3 underway, >50 total citations) on:

> A suite of novel on-chip sensors for detailed device aging measurements, validated via 12nm process implementations.

> Applied computational Bayesian inference of hierarchical, physics-based probabilistic models for lifespan prediction.

> Novel Bayesian experimental design objectives and algorithms for risk- and cost-aware qualification test optimization.

> Leveraging sensor telemetry for in-field reliability risk prediction (with an invited talk at the 2023 IEEE SLM workshop).

Industrial Experience

Capstone Teaching Assistant, University of British Columbia: *Vancouver, BC (Sept. 2020 – Present)*

> Provided technical guidance, mentorship, and evaluation for 25 student teams across diverse engineering projects.

> Advised on topics ranging from radar and acoustic sensors to machine learning models to software analytics processing.

Silicon Verification Intern, Microsoft Corporation: *Raleigh, NC (Sept. – Dec. 2019)*

> Designed dynamic SystemVerilog generation tooling, accelerating test simulations across four design teams.

> Led development of on-demand RTL memory generation system, eliminating vendor delays for digital designers.

ASIC Design Intern, NVIDIA Corporation: *Santa Clara, CA (Sept. – Dec. 2018)*

> Developed behavioural PCIe 4.0 clocking architecture RTL model for rapid design feedback and test simulations.

> Implemented and verified critical bug resolutions in ASIC graphics processors using Perl and Verilog.

Silicon Validation Engineering Intern, NVIDIA Corporation: *Santa Clara, CA (Jan. – Apr. 2018)*

> Developed on-demand pre-silicon power estimator, providing rapid chip binning product planning capabilities.

Hardware Designer, Tyco Security Products: *Vaughan, ON (May – Aug. 2017)*

> Designed and evaluated circuitry for cost-sensitive full-duplex wired communications in intrusion security products.

Software Developer, ADP Canada: *Mississauga, ON (Aug. – Dec. 2016)*

> Led developer workflow metrics initiative, streamlining data processing and analysis using Python and MongoDB.

Software Analyst, Thales Canada: *Vaughan, ON (Jan. – Apr. 2016)*

> Developed program state analysis tool using Excel VBA that aided in validation of SIL4 onboard train control systems.

Education

Candidate for Doctor of Philosophy in Electrical and Computer Engineering, University of British Columbia: *Vancouver, BC (Sept. 2020 – Present)*

Dissertation – Long-Term Reliability Monitoring and Qualification Strategies for Safety Critical Sub-Nanometre Semiconductors

Supervisor – Dr. André Ivanov **Awards** – NSERC Postgraduate Doctoral Scholarship (2023, $95,000)

Bachelor of Applied Science, Honours Co-operative Electrical Engineering, Management Sciences Option (with Dean's Honours List Distinction), University of Waterloo: *Waterloo, ON (Sept. 2015 – Apr. 2020)*

Awards – Savvas Chamberlain Scholarship (2020, $3,000) – Sandford Fleming Programming Competition Award (2019, $400)

Projects

12nm ASICs for Implementation of Novel Semiconductor Wear-Out Sensors: *(Aug. 2021 – Apr. 2023)*

> Sole designer of functional mixed-signal FinFET chip requiring highly complex simulation and verification workflows.

> Provided training, mentorship, and support to new designer of a second functional chip design with additional sensors.

Software Framework for Probabilistic Physics-of-Failure Model Simulation and Analysis: *(Jan. 2023 – Aug. 2025)*

> Designed engineering tool for computational Bayesian inference and experimental design using NumPyro and JAX JIT.

> Contains novel features and algorithms for non-expert use, multilevel variability, and risk-based experimental design.

Automated Semiconductor Accelerated Stress Testing System: *(Mar. 2023 – July. 2024)*

> Designed, built, and tested reliable hardware system using Rust firmware, capable of executing 1000-hour stress tests.

Intelligent Circuit Debugging via Bayesian Experimental Design: *(Oct. 2023 – Dec. 2023)*

> Designed tool for automated debug of linear analog circuits using custom probabilistic fault model and Pyro.

Personal Website Constructed Using Generative AI: *(Aug. 2025)*

> Steered Claude Code using detailed task planning to construct a personal website with minimal manual intervention.

Photonics Chip Design for Aging Analysis and Non-Invasive Monitoring: *(Jan. 2021 – Apr. 2021)*

> Sole designer of functional photonics chip implementing active ring resonators for monitoring device aging.

Technical Toolset

Hardware Design: Cadence Virtuoso Suite, KLayout, Verilog, VHDL, KiCAD, Altium, SPICE, Lumerical Suite

Software Design: Python (+NumPyro, Pyro, and PyMC), Rust, C, Perl, Java, MATLAB, Claude Code, Git, Perforce, MongoDB

Productivity: Windows (+WSL), Linux, Vim, LaTeX, Zotero, Microsoft Office

Audio/Visual Design: FL Studio, DaVinci Resolve, Affinity Publisher/Designer/Photo

Selected Publications

I. Hill and A. Ivanov, "Multilevel variability and epistemic uncertainty analysis of reliability physics models through hierarchical probabilistic modelling," in *2025 IEEE International Reliability Physics Symposium (IRPS)*, 2025, pp. 1-6.

I. Hill, M. Rendón, and A. Ivanov, "Enhanced wear-out sensor design in a 12nm process for separable stress regime monitoring," in *2024 IEEE 42nd VLSI Test Symposium (VTS)*, 2024, pp. 1-7.

I. Hill, M. Rendón, and A. Ivanov, "A novel induced offset voltage sensor for separable wear-out mechanism characterization in a 12nm FinFET process," in *2024 IEEE International Reliability Physics Symposium (IRPS)*, 2024, pp. 1-9.

I. Hill, P. Chanawala, R. Singh, S. A. Sheikholeslam, and A. Ivanov, "CMOS reliability from past to future: A survey of requirements, trends, and prediction methods," *IEEE Transactions on Device and Materials Reliability*, vol. 22, no. 1, pp. 1–18, 2022.

Interests

Classical Guitar *(2003 – Present)*: Lifelong guitarist at an advanced level, also playing electric guitar, trumpet, and voice.

Music Production *(2020 – Present)*: Self-taught capabilities in recording, mixing, mastering, and synth programming.

Recreational Sports *(2003 – Present)*: Pastime enjoyment of soccer, ultimate frisbee, downhill skiing, and sail racing.